

Isa Gutierrez

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EDUCATION

University of California, Berkeley

Expected Graduation: May 2028

B.S. Electrical Engineering and Computer Science

Relevant Coursework: Microelectronic Devices and Circuits, Computer Architecture, ASIC Digital Design, Microfabrication, Linear Circuits, Signals & Systems, Circuits & Devices, Data Structures, Linear Algebra

SKILLS

Design & Simulation: KiCad, OrCAD X, Altium, LTspice, Innovus, SolidWorks, SolidWorks Electrical, OnShape

Programming & Systems: C, C++, Verilog, LabVIEW, Python, Java, JavaScript, UNIX/Linux, Git, Wireshark

RELEVANT EXPERIENCE

Ti Lab @ UC Berkeley

Undergraduate Researcher

September 2026 – Present

- Researching DNA-origami photonic nanostructures using nanoscale self-assembly and semiconductor nanofabrication, with training in DNA-origami design and AFM characterization.

KLA

Electrical Design Engineering Intern

June 2026 – August 2026

- Designed a high-power (600+ W) evaluation PCB in OrCAD to evaluate high-power DC-DC converters, incorporating configurable I²C and PMBus interfaces to support converter communication and control.
- Designed and simulated multiple low-pass filter networks in LTspice for 30+ A power applications, achieving strong noise attenuation above 10 kHz for high-current semiconductor inspection electronics.
- Evaluated and characterized high-power DC-DC converters, then integrated converters into the broader power supply system powering high-speed imaging cameras used for semiconductor wafer inspection.

EUV Tech

Electrical Engineer Intern

June 2025 – January 2026

- Designed photoconductivity PCB using KiCad, multiplying max bias voltage 8x (from 25V to 200V) to improve response time and sensitivity of photodiode dark current and laser intensity measurements
- Developed photodiode emulator PCB to generate precise current pulses to tune, test, and calibrate preamps for utilization in extreme ultraviolet (EUV) lithography and metrology tools
- Engineered the complete electrical system of an automated high-vacuum chamber (1e-6 mbar) for residual gas analysis and semiconductor optics cleaning and executed a successful pumpdown test

Engineering Solutions at Berkeley

Technical Project Manager – Williams Group @ UC Berkeley

February 2026 – May 2026

- Managed end-to-end design of custom RF generator for quadrupole ion guide control, coordinating hardware, firmware, and PCB development from architecture through bench testing
- Oversaw implementation of a multi-stage architecture spanning waveform synthesizer, custom power PCB, and custom transformer to produce ± 1 kV differential waveforms across capacitive loads of 1 nF

Page Technologies Electrical Engineering Consultant

February 2025 – May 2025

- Engineered an automated hydroponic water sampling system utilizing custom PCB, ESP32, and C++ that enables 1-minute water quality analysis by seamlessly extracting, rerouting, and returning samples to pipe

PROJECTS

RISC-V ASIC Processor

- Designed and implemented a three-stage pipelined RISC-V processor core in SystemVerilog, developing synthesizable datapath and control logic, branch handling, memory interfaces, and pipeline control.
- Developed and integrated an FSM-based cache subsystem, verified functionality through simulation and benchmark testing, and completed the RTL-to-layout ASIC flow targeting the SkyWater 130 nm process.

Haptic Feedback Communication PCB | Neurotech @ Berkeley

- Developed a modular haptic feedback system, leading research and topology design for a low-noise, power-efficient architecture integrating an STM32-based PCB with barometric and magnetic sensors